# Structured notes: Section 1, electromotive force (Griffiths §7.1)

Formal notes. Griffiths, *Introduction to Electrodynamics* (5th ed.), §7.1 Electromotive force, pp. 298–315. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption.

## 7.1.1 Ohm's law

*Source: book-sources/m2-s71-ohm.html, Griffiths pp. 298–303. Equation numbers as printed in the book.*

### Prerequisites

- Current density $\mathbf{J}$; potential difference $V = -\int \mathbf{E}\cdot d\mathbf{l}$.
- Gauss's law; the divergence of a curl is zero; Laplace's equation.

**PLUS TWO RECALL: OHM'S LAW AND RESISTIVITY**

**Recall.** From Class 12: $V = IR$, and for a wire $R = \rho\ell/A$, where $\rho$ is the resistivity.

**Re-derive.**

1. Take a uniform cylinder of length $\ell$ and area $A$, with potential difference $V$ between its ends. The field is $E = V/\ell$ (uniform, Example 7.3).
2. The current density is $J = I/A$.
3. From $V = IR$ and $R = \rho\ell/A$: $I = VA/(\rho\ell)$.
4. Therefore $J = I/A = V/(\rho\ell) = E/\rho = \sigma E$, with $\sigma = 1/\rho$.

**Saved.** $\mathbf{J} = \sigma\mathbf{E}$ (Eq. 7.3).

### 1. Constitutive law

Let $\mathbf{f}$ be the force per unit charge that drives the charges. For most substances, $\mathbf{J}$ is proportional to $\mathbf{f}$:

$$\mathbf{J} = \sigma\mathbf{f} \qquad (7.1)$$

$\sigma$ is the conductivity; $\rho = 1/\sigma$ is the resistivity (Table 7.1). Metals: $\sigma = \infty$. Insulators: $\sigma \approx 0$.

For an electromagnetic drive, $\mathbf{f} = \mathbf{E} + \mathbf{v}\times\mathbf{B}$, so $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v}\times\mathbf{B})$ (Eq. 7.2). Neglecting the magnetic term:

**RESULT 1.1: OHM'S LAW**

$$\mathbf{J} = \sigma\mathbf{E} \qquad (7.3)$$

In a perfect conductor, $\mathbf{E} = \mathbf{J}/\sigma = \mathbf{0}$ even when a current flows, so connecting wires are equipotentials.

### 2. Resistance of a uniform cylinder (Ex. 7.1, Fig. 7.1)

For a cylinder of cross-sectional area $A$ and length $L$, with $\mathbf{E}$ uniform (Ex. 7.3), $\mathbf{J} = \sigma\mathbf{E}$ is uniform and

$$I = JA = \sigma EA = \frac{\sigma A}{L}V$$

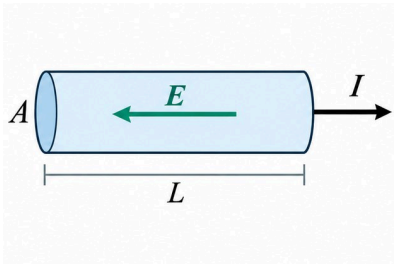



*Fig. 7.1 (book): cylindrical resistor, field E along the axis, current I leaving the right end.*

### 3. Resistance of coaxial cylinders (Ex. 7.2, Fig. 7.2)

Field between the cylinders:

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}}$$

where $\lambda$ is the charge per unit length of the inner cylinder. Then

- $I = \sigma\oint\mathbf{E}\cdot d\mathbf{a} = \frac{\sigma}{\epsilon_0}\lambda L$;

- $V = -\int_b^a \mathbf{E}\cdot d\mathbf{l} = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{b}{a}$.

Hence $I = \frac{2\pi\sigma L}{\ln(b/a)}V$.

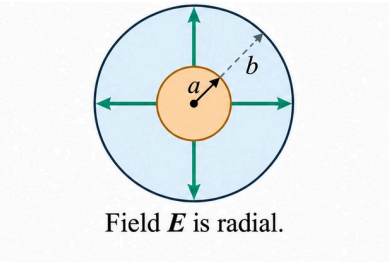



*Fig. 7.2 (book): coaxial cylinders, end view; the field E between them is radial.*

## 4. Circuit form

In both cases $I \propto V$:

> **RESULT 1.2: CIRCUIT FORM OF OHM'S LAW**
>
> $$V = IR \qquad (7.4)$$
>
> $R$ depends on geometry and $\sigma$. Ex. 7.1: $R = L/\sigma A$. Ex. 7.2: $R = \frac{\ln(b/a)}{2\pi\sigma L}$. Unit: $\Omega$ (1 Ω = 1 V/A).

## 5. Proof that the field in Ex. 7.1 is uniform (Ex. 7.3)

$V$ obeys Laplace's equation inside the cylinder. Boundary conditions: $V = 0$ at $z = 0$; $V = V_0$ at $z = L$; $\mathbf{J}\cdot\hat{\mathbf{n}} = 0$ on the curved surface, so $\partial V/\partial n = 0$ there. The function $V = V_0 z/L$ satisfies all of these. By the uniqueness theorem it is the solution. $\mathbf{E} = -\nabla V = -(V_0/L)\hat{\mathbf{z}}$ is uniform.

## 6. Steady current and charge density (Eq. 7.5)

$$\nabla\cdot\mathbf{E} = \frac{1}{\sigma}\nabla\cdot\mathbf{J} = 0 \qquad (7.5)$$

Charge density vanishes inside a homogeneous ohmic conductor carrying a steady current; unbalanced charge resides on the surface. Laplace's equation holds in such a medium.

> **PLUS TWO RECALL: JOULE HEATING**
>
> **Recall.** From Class 12: power $P = VI$, and with $V = IR$, $P = I^2R$.
>
> **Re-derive.**
>
> 1. Work per unit charge across a resistor is $V$.
> 2. Charge passing per unit time is $I$.
> 3. So energy per unit time is $VI$. Substituting $V = IR$ gives $I^2R$.
>
> **Saved.** $P = VI = I^2R$ (Eq. 7.7).

## 7. Joule heating (Eq. 7.7)

> **RESULT 1.3: JOULE HEATING LAW**
>
> $$P = VI = I^2R \qquad (7.7)$$
>
> $P$ in watts when $I$ is in amperes and $R$ in ohms.

## 8. Microscopic model (background, Eq. 7.6)

$$\mathbf{J} = nfq\,\mathbf{v}_{\text{ave}} = \left(\frac{nf\lambda q^2}{2mv_{\text{thermal}}}\right)\mathbf{E} \qquad (7.6)$$

*Classical (Drude) model with $n$ molecules per unit volume, $f$ free electrons per molecule, charge $q$, mass $m$, mean free path $\lambda$. Qualitative only; footnote 2 notes it is not the quantum theory.*

## 7.1.2 Electromotive force

*Source: book-sources/m2-s72-emf.html, Griffiths pp. 304–305.*

### Prerequisites

- Ohm's law and $V = IR$ (Section 7.1.1).
- For an electrostatic field, $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.

### 1. Current continuity in a circuit

In a steady circuit the current is the same at all points of the loop. An imbalance produces charge accumulation, whose electric field opposes the imbalance (Figs. 7.7, 7.8). The process is quick enough that the current is effectively uniform even at radio frequencies.

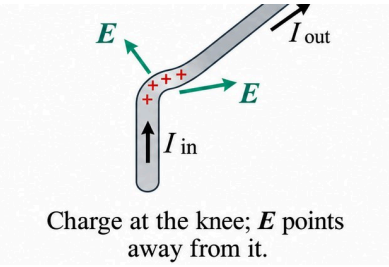



*Fig. 7.8 (book): charge piles up at a bend, and E points away from the knee.*

### 2. Force decomposition

The total force per unit charge splits into a source part and an electrostatic part:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E} \tag{7.8}$$

$\mathbf{f}_s$ is non-zero only in the source region. $\mathbf{E}$ acts throughout the circuit.

**PLUS TWO RECALL: EMF AS WORK PER UNIT CHARGE**

**Recall.** From Class 12: the emf of a source is the work done per unit charge by the source in carrying charge round the circuit.

**Re-derive.**

1. The source force per unit charge is $\mathbf{f}_s$. Its work per unit charge over one trip round the loop is $\oint \mathbf{f}_s \cdot d\mathbf{l}$.
2. The electrostatic field does no net work round a closed loop, since $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.
3. So only the source force remains.

**Saved.** $\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}$ (Eq. 7.9).

### 3. Definition of emf

**RESULT 1.4: ELECTROMOTIVE FORCE**

$$\mathcal{E} \equiv \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} \tag{7.9}$$

Equality holds because $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for electrostatic fields. Emf has the unit of volts; it is an integral of force per unit charge, not a force.

### 4. Ideal source

For an ideal source ($\sigma = \infty$), $\mathbf{E} = -\mathbf{f}_s$ within the source. The terminal potential difference is

$$V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E} \tag{7.10}$$

The integral extends over the loop because $\mathbf{f}_s = \mathbf{0}$ outside the source. A real source has internal resistance $r$ and terminal voltage $\mathcal{E} - Ir$ (footnote 4).

**PLUS TWO RECALL: INTERNAL RESISTANCE**

**Recall.** From Class 12: a real cell of emf $\mathcal{E}$ and internal resistance $r$ gives terminal voltage $V = \mathcal{E} - Ir$ when current $I$ flows.

**Re-derive.**

1. An ideal source (no resistance) gives terminal voltage $V = \mathcal{E}$ (Eq. 7.10).
2. In a real source the current $I$ flows through the internal resistance $r$, so there is a potential drop $Ir$ inside the source (Ohm's law across $r$).
3. The terminal voltage is the emf less this drop.

**Saved.** $V_{\text{terminal}} = \mathcal{E} - Ir$ (footnote 4).

### 5. Hydraulic analogy (footnote 5)

Gravity corresponds to $\mathbf{E}$, the pump to the source, and height to voltage.

## 7.1.3 Motional emf and the flux rule

*Source: book-sources/m2-s73-motional.html, Griffiths pp. 306–315. Problems not included.*

### Prerequisites

- Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$; magnetic flux $\Phi = \int \mathbf{B} \cdot d\mathbf{a}$.
- Scalar triple product; Stokes' theorem; divergence.

### 1. Motional emf in a rectangular loop

Setup (Fig. 7.10): loop side $ab$ of length $h$ lies in a uniform field $\mathbf{B}$ into the page; the loop moves to the right at speed $v$.

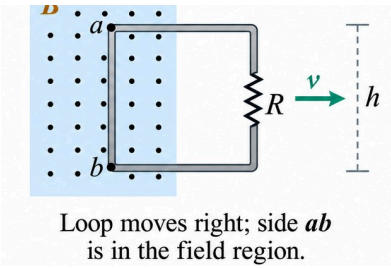



*Fig. 7.10 (book): rectangular loop with one side in a uniform field B; pulled to the right at speed v.*

The magnetic force per unit charge on the moving side is $\mathbf{f}_{\text{mag}} = \mathbf{v} \times \mathbf{B}$, with vertical component $vB$. Only side $ab$ contributes.

**RESULT 5: MOTIONAL EMF**

$$\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh \qquad (7.11)$$

The integral is evaluated at one instant (a snapshot of the loop). Sides $bc$ and $ad$ contribute zero.

**PLUS TWO RECALL: MOTIONAL EMF $\mathcal{E} = Blv$**

**Recall.** From Class 12: a conductor of length $l$ moving at speed $v$ perpendicular to a field $B$ has emf $Blv$.

**Re-derive.**

1. The magnetic force on a charge $q$ is $q\mathbf{v} \times \mathbf{B}$. So the force per unit charge is $\mathbf{v} \times \mathbf{B}$, with magnitude $vB$, directed along the wire.
2. Integrate this along the moving side of length $h$: $\oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh$.

**Saved.** $\mathcal{E} = vBh$ (Eq. 7.11).

### 2. Energy balance

The magnetic force does no work on a charge. Let $\mathbf{u}$ be the drift velocity along the wire and $\mathbf{w} = \mathbf{v} + \mathbf{u}$ the resultant velocity. The magnetic force $q\mathbf{w} \times \mathbf{B}$ has a component $quB$ opposing the drift. An external agent exerts $f_{\text{pull}} = uB$ to sustain the current (Fig. 7.11). The work per unit charge along the actual path, of length $h/\cos\theta$, is

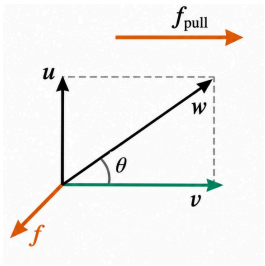



*Fig. 7.11 (book): velocity and force vectors for a charge in the moving wire.*

$$\int \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} = (uB)\left(\frac{h}{\cos\theta}\right)\sin\theta = vBh = \mathcal{E}$$

The emf is obtained along a different path from the work (Fig. 7.12). The pulling force does the work but contributes nothing to the emf; the magnetic force contributes to the emf but does no work.

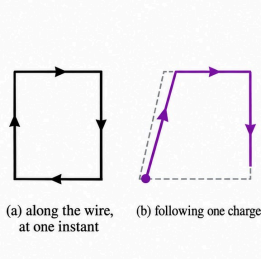



*Fig. 7.12 (book): (a) path for the emf, at one instant; (b) path followed by one charge.*

## 3. Flux rule

$$\Phi \equiv \int \mathbf{B} \cdot d\mathbf{a} \tag{7.12}$$

For the loop of Fig. 7.10, $\Phi = Bhx$ and $d\Phi/dt = -Bhv$ (since $dx/dt < 0$). The emf equals minus the rate of change of flux:

**RESULT 6: FLUX RULE**

$$\mathcal{E} = -\frac{d\Phi}{dt} \tag{7.13}$$

Valid for arbitrary loop shape, motion and field non-uniformity.

**PLUS TWO RECALL: FARADAY'S LAW: $\mathcal{E} = -d\Phi/dt$**

**Recall.** From Class 12: the induced emf equals minus the rate of change of magnetic flux, $\mathcal{E} = -d\Phi/dt$.

**Re-derive.**

1. For the loop of Fig. 7.10, the flux is $\Phi = Bhx$, where $x$ is the length of the loop inside the field.
2. As the loop moves right, $x$ decreases at speed $v$, so $dx/dt = -v$.
3. Therefore $d\Phi/dt = -Bhv$, and $-d\Phi/dt = vBh$, which is the motional emf of Eq. 7.11.

**Saved.** $\mathcal{E} = -d\Phi/dt$ (Eq. 7.13).

## 4. Proof

Let $\mathcal{S}$ and $\mathcal{S}'$ be surfaces bounded by the loop at $t$ and $t + dt$. The change in flux is the flux through the ribbon joining the two loops. A point $P$ on the loop moves to $P'$; the ribbon element is $d\mathbf{a} = (\mathbf{v} \times d\mathbf{l})\, dt$. Hence

$$\frac{d\Phi}{dt} = \oint \mathbf{B} \cdot (\mathbf{v} \times d\mathbf{l}) = \oint \mathbf{B} \cdot (\mathbf{w} \times d\mathbf{l})$$

$$= -\oint (\mathbf{w} \times \mathbf{B}) \cdot d\mathbf{l} = -\oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = -\mathcal{E}$$

The second equality uses $\mathbf{u} \parallel d\mathbf{l}$; the third uses the scalar triple product.

**Sign convention.** The direction of $d\mathbf{l}$ and $d\mathbf{a}$ are linked by the right-hand rule. A negative emf indicates current flowing in the negative sense.

## 5. Limitation: flux-rule paradox (Fig. 7.14)

The flux rule requires a single current path. Throwing the switch from $a$ to $b$ doubles the flux through the circuit but produces no motional emf; the ammeter reads zero. Switches, sliding contacts and extended conductors violate the single-path assumption.

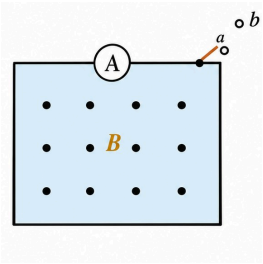



*Fig. 7.14 (book): flux-rule paradox; the switch moves from a to b.*

## 6. Example 7.4: Faraday disk

Disk of radius $a$ with angular velocity $\omega$ in a field $\mathbf{B}$ along $\hat{\mathbf{z}}$ (Fig. 7.15).

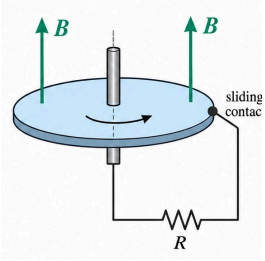



*Fig. 7.15 (book): Faraday disk spinning in a field B pointing up.*

$$\mathcal{E} = \int_0^a \omega s B\, ds = \frac{\omega B a^2}{2}, \qquad I = \frac{\omega B a^2}{2R}$$

The flux rule is not applicable in its single-path form, because the current spreads across the disk.

## 7. Eddy currents: Example 7.5

Spherical shell of conductivity $\sigma$, radii $R_1$, $R_2$, angular velocity $\boldsymbol{\omega}$, uniform $\mathbf{B}$. Current law: $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ (Eq. 7.2). Taking the divergence and using Gauss's law and continuity, in the steady state:

$$\rho = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B}) \qquad (7.14)$$

With $\mathbf{B} = B\hat{\mathbf{z}}$, $\boldsymbol{\omega} = \omega\hat{\mathbf{z}}$ and the BAC–CAB rule:

$$\mathbf{v} \times \mathbf{B} = \omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}) \qquad (7.15)$$

Hence $\rho = -2\epsilon_0\omega B$ (Eq. 7.16) and $\nabla^2 V = 2\omega B$ in the shell (Eq. 7.17), with compensating surface charge at the boundaries.

### CASE (A): $\boldsymbol{\omega} \parallel \mathbf{B}$

Radial current vanishes at $R_1$, $R_2$ (Eq. 7.18). Solution $V = \frac{\omega B}{2}(x^2 + y^2) + K$ (Eq. 7.21), so $\mathbf{E} = -\mathbf{v} \times \mathbf{B}$ (Eq. 7.22) and $\mathbf{J} = \mathbf{0}$.

$$E_r = -\omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}) \cdot \hat{\mathbf{r}} = -\omega B r \sin^2\theta \quad (7.19)$$

$$\left.\frac{\partial V}{\partial r}\right|_{R_1} = \omega B R_1 \sin^2\theta, \qquad \left.\frac{\partial V}{\partial r}\right|_{R_2} = \omega B R_2 \sin^2\theta \quad (7.20)$$

$$V(r,\theta) = \frac{\omega B}{2} r^2 \sin^2\theta + K \qquad (7.21)$$

### CASE (B): $\boldsymbol{\omega} \perp \mathbf{B}$

$\boldsymbol{\omega} = \omega\hat{\mathbf{x}}$, $\mathbf{B} = B\hat{\mathbf{y}}$: $\rho = 0$, $\nabla^2 V = 0$ in the shell, with $\partial V/\partial r$ specified at $R_1$ and $R_2$. Solution $V = -\frac{\omega B}{2} xy + K$. The eddy current $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ is non-zero.

*Footnote 10: the convection current is smaller than the conduction current by a factor of order $\epsilon_0\omega/\sigma$.*